

Bohao (Lexi) Yang

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Incoming Phd Student in Quantitative Biology & Medicine in Duke-NUS Medical School

EDUCATION

Duke University

Aug 2024 – May 2026

M.S. in Statistical Science | GPA: 3.74/4.0

Durham, NC, United States

- **Relevant Courses:** Multilevel & Hierarchical Models, High-Dimensional Statistics, Probabilistic Machine Learning, Natural Language Processing, Statistical Inference, Bayesian Statistical Modeling, Data Analysis at Scale in Cloud

University of Liverpool

Sep 2020 – Jun 2024

B.S. in Mathematics & Finance | GPA: 3.94/4.0 (First Class)

Liverpool, United Kingdom

- **Relevant Courses:** Stochastic Modeling, Real Analysis, Linear Statistical Models, Numerical Methods, Financial Mathematics, Operational Research, Metric Spaces, Linear Algebra, Calculus, Multivariable Calculus

MANUSCRIPTS

- **Yang B, Sun K, Pollak K, Pencina M, Hong C.** An Agentic AI Framework for Multi-Framework Communication Coding. *Npj Digital Medicine (under review); preprint*
- Zhuhan B, **Yang B**, Joshua Biro, Anand Chowdhury, Sun K, Pollak K, Pencina M, Hong C. An AI-Driven Framework for Generating High-Fidelity Clinical Dialogues to Evaluate Digital Scribing Systems. **In preparation**

RESEARCH EXPERIENCE

Multi-site ICU Electronic Health Record (EHR) Analysis for Equitable Mortality Prediction

Nov 2025 – Present

Advised by Drs. Chuan Hong, Deepshikha Ashana, and Matthew Engelhard et al.

Duke University

- An early-stage research project aimed at understanding and mitigating racial and ethnic bias in ICU mortality risk prediction, with the long-term goal of developing equitable, clinically deployable prognostic models.
- Preprocessed large-scale EHR data from Duke University and the Medical University of South Carolina and assisted with cohort construction.
- Assisted in the evaluation of existing mortality prediction models (e.g., SOFA, APACHE II, and time-to-event models such as Cox proportional hazards), focusing on robustness and fairness.

SAGE: Structured Agent for Diagnosis-Based EHR Cohort Extraction

Oct 2025 – Present

Advised by Drs. Chuan Hong and Tianxi Cai (with Harvard-affiliated investigators)

Duke University

- Contributed to a modular, SQL-based agent (SAGE) for clinical cohort identification from EHR data using diagnosis codes (ICD-9/10), and mentored junior students on implementation.
- Assisted with cohort extraction pipelines for multi-source EHR data, including diagnosis date identification and longitudinal incidence tracking.
- Supported testing and documentation of reusable phenotyping logic for inflammatory bowel disease (IBD; UC/CD), with extensibility to additional disease phenotypes.

MOSAIC: An Agentic AI for Multi-Framework Communication Analysis

Jan 2025 – Present

Advised by Dr. Chuan Hong; collaborated with Drs. Michael Pencina, Kai Sun and Kathryn Pollak

Duke University

- Developed a novel multi-agent framework based on LangGraph and Retrieval Augmented Generation (RAG), focusing on multi-scale analysis of clinician-patient dialogues.
- Designed a human-in-the-loop dynamic few-shot prompting strategy and domain-adaptive retrieval; trained and evaluated on 70 transcripts (~20 hours of audio), boosting human alignment to 93%+ and F1 to 90%+.
- Integrated locally deployed large language model (LLM) and Text-to-Speech (TTS) model with frequency-guided prompting to generate aligned multimodal text-audio pairs.
- Compared audio feature analysis methods (e.g., embeddings, emotion recognition, and acoustic event detection) and explored extensions on our current evaluation framework.
- Authored a manuscript detailing the methods, ablation studies, multi-dimensional evaluation (accuracy, fairness and robustness) and results.
- Co-authored a manuscript in ablation studies and evaluation methods sections; also contributed to the comparison analyses across different language models and TTS models.

An Internal Ambient Digital Scribing (ADS) System

Jan 2025 – Present

Advised by Dr. Chuan Hong; collaborated with Drs. Michael Pencina, Kai Sun and Kathryn Pollak

Duke University

- Developed an end-to-end privacy-safe ambient scribing tool, integrating voice activity detection (VAD), Whisper-based Automatic Speech Recognition (ASR), and voice embeddings-powered diarization.
- Utilized the internal ADS tool to generate structured medical notes for clinicians on a weekly basis, effectively reducing manual scribing time and documentation burden.
- Implemented human-in-the-loop evaluation with clinicians to refine factual consistency and transcribing accuracy.

Deep Learning for Electroencephalogram (EEG) in Mental Health Disorder Diagnostics

Jun 2023 – Dec 2023

Advised by Dr. Feng Liu

Stevens Institute of Technology

- Investigated statistical, machine learning, and deep learning models for EEG-based mental health disorder diagnosis,

emphasizing model interpretability and diagnostic reliability.

- Conducted comparative experiments across convolutional neural networks (CNNs), recurrent neural networks (RNNs), and graph neural networks (GNNs) to evaluate spatial-temporal features and brain connectivity patterns.
- Authored a literature review summarizing state-of-the-art neural network architectures for EEG-based mental health disorder diagnosis and prognosis.

Quantitative Analysis of Carbon Trading and Sustainable Finance

Jun 2022 – Sep 2022

Advised by Wen Qian

Xi'an Jiaotong Liverpool University

- Analyzed China's carbon market using emissions trajectories and allowance price dynamics, examining lagged correlations and cross-sector co-movements and interpreted policy-driven market dynamics
- Modeled structural transition risk by linking carbon-intensity trends with financial data using clustering and regression.
- Delivered a research report with data-driven recommendations for financial institutions.

Machine Learning Model for Credit Default Prediction

Dec 2021 – Mar 2022

Advised by Dr. Zhen Ye

Imperial College London (Winter School)

- Cleaned and visualized the real-world credit data using Python.
- Trained and benchmarked multiple classifiers (e.g., XGBoost, Random Forest, Logistic Regression).
- Deployed the final XGBoost (AUC = 0.96, Precision = 0.93) as a FastAPI + Docker for scalable credit-risk scoring.

PROFESSIONAL EXPERIENCE

ByteDance

May 2025 – Aug 2025

AI Product Manager Intern

China

- Diagnosed conversion bottlenecks in drug-related searches via funnel analysis and user data analysis.
- Optimized the agent workflow of “Xiaohe” AI Medical App (from a single-agent to an SFT-enhanced multi-agent), improving content quality and user experience and driving GMV growth.
- Reduced hallucination and improved the retrieval strategy by refining drug-knowledge RAG and prompt engineering.

LoopLand

Dec 2024 – Apr 2025

AI Product Manager Intern

Remote

- Participated in 0-1 product design of an AI music creation platform for B-end and Pro-C users.
- Participated in pre-training a music feature extraction model and designing a loop + natural language dual-control music generation framework based on U-shaped Vision Transformer and a time window editing system.
- Conducted user interviews to identify core inefficiencies, performed competitive analysis on global AI music products, and produced a comparative matrix of key products.

Bank of China

Jun 2023 – Aug 2023

Risk Modelling Intern

China

- Integrated credit records via SQL and analyzed delinquency and guarantee risk factors using multivariate methods.
- Trained logistic regression and XGBoost models for credit scoring and default risk prediction.
- Conducted stress testing via Monte Carlo simulation and assessed solvency risks to support post-loan management.

TianFeng Securities

Mar 2022 – May 2022

Data Analyst Intern

China

- Conducted data analysis and visualization via Python and delivered insights on financial information.
- Assisted with desk research and organized due diligence files and internal briefing materials.
- Analyzed core financial metrics (net profit, ROE, EBITDA) to evaluate issuer solvency and bond pricing.

TEACHING EXPERIENCE

DEC 546Q – Modern Analytics

Oct 2025 – Dec 2025

Teaching Assistant

Duke University, Fuqua School of Business

- Graduate-level course covering machine learning and AI topics, including neural networks, diffusion models and LLM.
- Held weekly office hours to support students with model training, regularization, and evaluation in Python.
- Graded on students' coding assignments, case studies and final projects, providing structured feedback.

HONORS & AWARDS

- Sampford Memorial Prize (The best final-year student in Statistics & Operational Research) Jul 2024
- University Exchange Scholarship (Top 1%) Jul 2023
- University Academic Excellence Award (Top 1%) Jun 2021, Jun 2022

ADDITIONAL INFORMATION

- **Technical Skills:** Python (PyTorch), R, SQL, Rust, MATLAB, Maple, Docker, Git, AWS
- **Languages:** English (Proficient), Mandarin (Native), Spanish (Conversational)
- **Interests:** Climbing, Badminton, Guitar, Reading